

230162



UA-7447

Integrated M. C. A. (Sem. VI)
Examination, June - 2025

IMCA-601T

Computer Graphics

Time allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

Note : This question paper contains two sections as under :

Section-A**Max. Marks - 20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

SECTION - A

- 1 (i) What are Random-Scan displays?
- (ii) Write down the application fields of computer graphics.
- (iii) Write the properties of circles.
- (iv) What is polygon filling in computer graphics?
- (v) What is composite transformation? Provide an example.
- (vi) What is the effect of scanning on an object's size and orientation in 3D?
- (vii) Define parallel projection.
- (viii) How do you calculate vanishing points for a given line or plane?
- (ix) What is the CMYK color model used for?
- (x) How does the HSV color model differ from RGB?

10×02=20**[Contd...****SECTION - B****UNIT - I**

- 2 Describe about any four graphical input devices. 10

OR

Describe the architecture of simple Raster graphics system.

10**UNIT - II**

- 3 Derive the midpoint circle algorithm and explain its steps. Provide an example to illustrate its application. 10

OR

Explain how to draw thick primitives using the Bresenham's line algorithm. Discuss the challenges and limitations of this approach.

10**UNIT - III**

- 4 Discuss briefly about 2D composite transformations. Also discuss about 2D reflection and 2D shear. 10

OR**UA-7447]****3****[Contd...**

Derive the 3D transformation matrix for rotation about an arbitrary axis. Explain the steps involved.

10

UNIT - IV

- 5** Discuss the importance of visible surface determination in computer graphics and its applications in real time rendering and animation.

10

OR

Explain the Z-buffer algorithm for hidden surface elimination. Discuss its advantages and disadvantages.

10

UNIT - V

- 6** Compare and contrast 2D and 3D animation techniques. Discuss their application in film, television, and gaming.

10

OR

Discuss the importance of color models in computer graphics and digital design.

10